Quick rules and their traps

Optimization · Level 1 · Lesson 3



Today we will

- Make a plan fast with a quick rule.
- Improve the plan with small swaps.
- Ask who pays for a rule, not only the total.

Remember

- Quick rules (nearest first, most per kilo first) are fast and often good, but promise nothing.
- Small swaps that remove crossings improve a round; keep only the ones that help.
- Trying every plan explodes: 9 villages make 181,440 rounds, 20 more than a billion billion.

A human bus route

1. Warm-up (5 min): How does our bus driver choose the order?
2. Teach (8 min): Nearest first, and small swaps.
3. Bus round (15 min): A pupil “bus” walks nearest first.
4. Swap (7 min): Find crossings, swap, count again.
5. Wrap-up (5 min): Who rode the longest?

Spread the “villages” unevenly, with a cluster on one side and two far away, so nearest first leaves long legs at the end. A ball of string makes crossings easy to see.

Let’s discuss

1. Why are the last legs of a nearest-first round so long?
2. Why not just try every possible round?
3. Who pays for the nearest-first rule, and how could we share that fairly?

Exit ticket

- Name one quick rule we use at school, one time it could fail, and how we could check it.